

Miles Franck

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SUMMARY

Software engineer with a technical foundation in Python, database systems, and systems-level programming in Rust, including concurrent and async application design. Experienced building backend architectures, debugging complex systems, and collaborating across technical and non-technical stakeholders. Brings a security-aware development mindset from SOC analyst experience, with CompTIA Security+ in progress. Looking to grow within a collaborative engineering team building reliable, well-architected software.

CORE COMPETENCIES

Software Design & Architecture • Concurrent / Async Programming • Database Design • Debugging & Troubleshooting • Secure SDLC • Agile / Scrum Collaboration • Technical Documentation

TECHNICAL SKILLS

Languages & Databases: Python, Rust, PostgreSQL

Systems & Tooling: Git, WSL, Tokio (async runtime)

Security Tools (Familiar): Shodan, Nmap, Splunk, Wireshark

EXPERIENCE

Summer Fellow — Carnegie Mellon University, Heinz College Summer Fellowship *2025*

- Completed a hands-on technical fellowship applying software engineering and systems thinking to real-world technical challenges.
- Applied technical problem-solving skills in team-based exercises and collaborative development projects.
- Collaborated with industry professionals on system architecture, technical solutions, and project management initiatives.

SOC Operations Analyst II — BCR Cyber *Dec 2025*

- Participated in a live-fire cyber range exercise, identifying and tracing an active attacker's presence within a simulated enterprise environment

SOC Operations Analyst I — BCR Cyber *Nov 2025*

- Applied technical troubleshooting and analytical skills within security operations workflows, including log analysis, alert triage, and structured incident documentation.

Teaching Assistant — Allegheny College *2022 – 2025*

- Provided one-on-one technical support and debugging help to students during office hours, translating complex concepts into clear, actionable guidance.
- Reviewed student code and projects for correctness, structure, and adherence to best-practice coding standards.
- Communicated across technical and non-technical audiences to align grading criteria and project expectations with faculty.

PROJECTS

Rust TCP Chat Server *2026*

- Built a multi-client TCP chat server in Rust using the Tokio asynchronous runtime, supporting concurrent client connections and real-time message broadcasting.
- Implemented a publish-subscribe messaging system using Tokio broadcast channels to efficiently distribute messages to all connected clients.
- Designed a room management system ("salts") with unique join codes, user authentication, connection limits, and participant tracking.

BanjoDB *2025*

- Built a dual-language architecture separating a Rust core from a Python interface to emulate enterprise-style backend isolation.

- Implemented Argon2 password hashing with unique per-user salts as part of the application's authentication layer.

Automated Vulnerability Analysis Tool *2024*

- Built a Python tool that queries the Shodan API to enumerate internet-facing assets and cross-references results against CVE/CPE vulnerability databases.

Deepseeking-Vulns *2025*

- Developed an AI-assisted analysis workflow integrating Shodan, Ollama, and DeepSeek to process and evaluate exposed network service data.

CERTIFICATIONS

- CompTIA Security+ — CompTIA (Expected October 2026)
- SOC Operations Analyst II — BCR Cyber (Dec 2025)
- SOC Operations Analyst II — BCR Cyber (Nov 2025)